

MRCT Consistency Evaluation via Subgroup Identification

A ForestSearch Framework with a View Towards Biomarker Causal Effects

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- ① **Regional Consistency:** The MRCT regional consistency challenge
- ② **Methodology:** Weibull AFT / Cox model with spline biomarker effects
- ③ **Estimands:** Marginal HR, Average HR, and Controlled Direct Effect
- ④ **ForestSearch Algorithm:** Exploratory subgroup identification
- ⑤ **Case Study:** Synthetic MRCT with biomarker-driven treatment heterogeneity
- ⑥ **Simulation Design:** DGM specification and per-replicate workflow
- ⑦ **Operating Characteristics:** HTE and Null scenarios
- ⑧ **Summary and Discussion**

Exploratory Subgroup Identification Analyses

The Scenario

- A **global oncology trial** meets statistical significance for the ITT population
- **Asia-Pacific (AP)** comprises 15–20% of the trial
- **Consistency criterion:** positive point estimate (Cox model $HR < 1.0$) in AP region
- **Problem:** AP regional HR estimate ≈ 1.0 — consistency appears to fail

Key Question

Can a subgroup $\mathcal{G}$ with **enhanced benefit** be identified in non-AP such that $AP(\mathcal{G})$ meets regional consistency?

Forest Search (León et al., 2024)

Imagine a forest plot evaluating all-possible combinations of covariates (e.g. $ecog = 0$, $ecog = 1$, bm low, bm high, $ecog = 0$ & bm low, etc)

Signals of harm (Cox estimate > 1.25 threshold); *Reproducibility?*

Weibull AFT / Cox Dual Representation

AFT Model (survreg):

$$\log(T) = \log(\theta) + L'\gamma + \tau\varepsilon$$

where $\tau = 1/\nu$ and $\varepsilon \sim$ extreme value.

Cox Hazard Model:

$$\lambda(t; L) = \lambda_0(t) \exp(L'\beta)$$

with Weibull baseline $\lambda_0(t) = \nu\theta^{-\nu}t^{\nu-1}$.

AFT-to-Hazard Transformation

$$\beta = -\gamma / \tau$$

R's survreg estimates $(\log \hat{\theta}, \hat{\gamma}, \hat{\tau})$ on the AFT scale; hazard-ratio coefficients are obtained by dividing by the scale parameter.

Potential outcomes are *simulated* on the AFT scale while treatment effects are *interpreted* on the hazard-ratio scale.

Two-Phase Spline Model

Cox linear predictor with biomarker Z and treatment X :

$$L'\beta := \beta_1 X + \beta_2 Z + \beta_3 ZX + \beta_4 (Z - k)\mathbf{1}(Z > k) + \beta_5 (Z - k)\mathbf{1}(Z > k)X$$

Parameter	Term	Role
β_1	X	Main treatment effect (intercept of log-HR)
β_2	Z	Prognostic biomarker effect (both arms)
β_3	ZX	Biomarker $\times$ treatment interaction ($Z \leq k$)
β_4	$(Z - k)\mathbf{1}(Z > k)$	Spline: prognostic slope change above k
β_5	$(Z - k)\mathbf{1}(Z > k)X$	Spline: treatment-effect slope change above k

Causal Log-HR and Spline Parameterization

Causal log-hazard-ratio for biomarker level z :

$$\psi^0(z) = \begin{cases} \beta_1^0 + \beta_3^0 z, & z \leq k \\ \beta_1^0 + \beta_3^0 z + \beta_5^0(z - k), & z > k \end{cases}$$

Spline Parameterization:

$$\beta_1^0 = \psi^0(0)$$

$$\beta_3^0 = \frac{\psi^0(k) - \psi^0(0)}{k}$$

$$\beta_5^0 = \frac{\psi^0(\zeta) - \psi^0(0) - \beta_3^0 \zeta}{\zeta - k}$$

Working Example:

z	$\psi^0(z)$	HR(z)
0	$\log(3.00)$	3.00
5 (knot)	$\log(1.25)$	1.25
10 (ζ)	$\log(0.50)$	0.50

Harm at low z , crossover near knot,
benefit at high z .

Potential Outcomes and Estimands

Potential Outcome (PO) $T[x, z]$, under treatment condition x (set to $x = 1[0]$ for treatment[control]) and biomarker level z

PO contrast

$\psi^0(z) = \{m(0, z) - m(1, z)\} / \tau$: Difference in true means of POs $\log(T[0, z]) - \log(T[1, z])$.

Strong prognostic W : Linear predictor

$$I_{x,z,w} = \beta_1^0 x + \beta_2^0 z + \beta_3^0 z x + \beta_4^0 (z - k) I(z > k) + \beta_5^0 (z - k) I(z > k) x + \beta_w^0 w$$

PO hazard function (omitting baseline)

$$\theta^x(z, v) = \exp(I(x, z, v))$$

Average Hazard Ratio (AHR)

$$\text{AHR}(z^+) = \exp\{E_{Z \geq z} \psi^0(Z)\}$$

Exponentiated average of differences in true means of POs (causal log-hazard ratios across z)

Controlled Direct Effect (CDE)

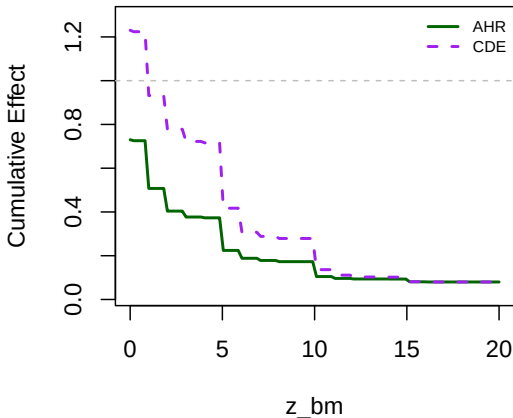
$$\text{Let } \bar{\theta}^x(z+) = E_{Z \geq z, V} \theta^x(Z, V)$$

$$\text{CDE}(z^+) = \bar{\theta}^1(z+) / \bar{\theta}^0(z+)$$

Ratio of average exponentiated hazards

Three Treatment Effect Estimands

```
dgm_spline <- generate_aft_dgm_flex(  
  data = df.case,  
  continuous_vars = c("age", "bm"),  
  factor_vars = c("male", "histology",  
                  "prior_treat", "AP"),  
  set_beta_spec = list(set_var = c("z_AP"),  
                       beta_var = -log(5)),  
  continuous_vars_cens = c("age"),  
  factor_vars_cens = c("prior_treat"),  
  cens_type = "weibull",  
  outcome_var = "tte", event_var = "event",  
  treatment_var = "treat",  
  subgroup_vars = NULL, subgroup_cuts = NULL,  
  k_inter = 0.0, model = "alt",  
  spline_spec = list(  
    var = "z_bm", knot = 5, zeta = 10,  
    log_hrs = log(c(3, 1.25, 0.5))  
  ),  
  verbose = FALSE, standardize = FALSE  
)
```



Subgroup Defined by (general) H

AFT Model Specification

$$\log T_i(x=1) = \mu + \mathbf{L}_i(1)^\top \gamma + \tau \varepsilon_i$$

The covariate vector includes treatment, prognostic effects (V , say), and an interaction:

$$\mathbf{L}_i(x=a)^\top \gamma = a \cdot \gamma_{\text{treat}} + \mathbf{V}_i^\top \gamma_V + a \cdot H_i \cdot \gamma_{\text{int}}$$

where $H_i \in \{0, 1\}$ flags harm-subgroup membership ($\mathcal{S}$).

$$\text{AHR}(\mathcal{S}) = \exp \left\{ \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \psi_i^0 \right\}$$

The **marginal hazard ratio** for a subgroup $\mathcal{S}$ is obtained from a Cox model fit to stacked potential outcomes:

$$\text{HR}_{\text{marg}}(\mathcal{S}) = \exp(\hat{\beta}_{\text{Cox}} \mid \text{data} = \{(T_i(1), 1, 1), (T_i(0), 1, 0) : i \in \mathcal{S}\})$$

Computation in ForestSearch:

- 1 Generate shared error: $\varepsilon_i = \log(\text{Exp}(1))$
- 2 Compute potential survival times:

$$\log T_i(1) = \mu + \sigma \varepsilon_i + \mathbf{L}_i(1)^\top \gamma$$

$$\log T_i(0) = \mu + \sigma \varepsilon_i + \mathbf{L}_i(0)^\top \gamma$$

- 3 Fit: $\text{HR}_{\text{marg}} = \exp(\hat{\beta}_{\text{treat}})$

ForestSearch Algorithm (León et al., 2024, *SIM*)

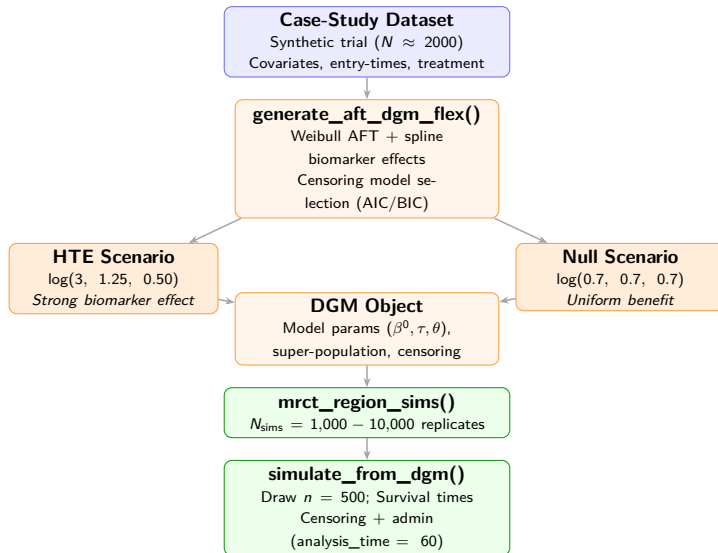
- ❶ **Candidate Selection** — From baseline covariates to L candidate subgroup indicators
 - *LASSO*: Cox regularization for prognostic selection; mitigates false discovery from noise
 - *GRF*: Causal survival forest targets RMST to identify treatment effect heterogeneity
 - Continuous covariates $\rightarrow$ binary splits (medians, quartiles, GRF cuts)
- ❷ **Exhaustive Subgroup Search** — All combos of L candidates (default max of 2-way)
 - $L(L-1)/2 + L$ possible combinations from L subgroup indicators (max = 2)
 - Filter: $n \geq 60$, events $d_0 \geq 12$, and $d_1 \geq 12$
 - Screen: Cox HR ≥ 1.25 (experimental vs. control) “signal of harm”
- ❸ **Splitting Consistency** — Validate each candidate ($R = 1,000$ splits)
 - Randomly split subgroup 50/50; require *both halves* HR ≥ 1.0 “consistent with harm”
 - $\hat{p}_{\text{cons}} = R^{-1} \sum_r I(\min(\hat{\beta}_s^{1r}, \hat{\beta}_s^{2r}) \geq 0) \geq 90\%$
 - Select $\hat{H}$: subgroup “maximally consistent with harm”

Three Subgroup Identification Objectives

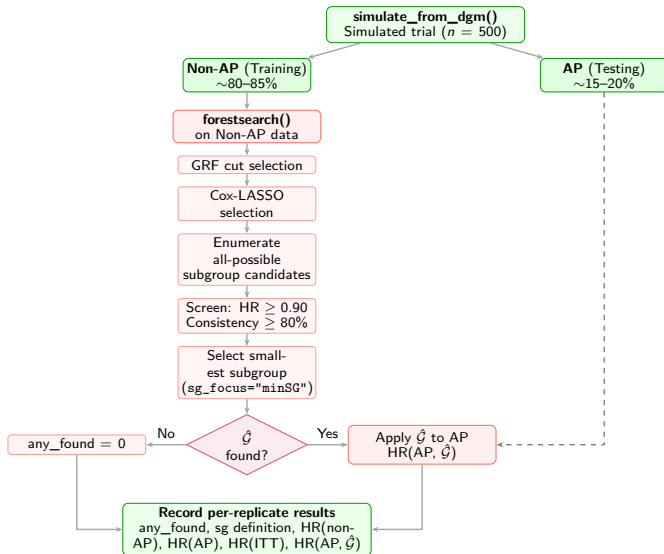
Objective	Goal	Criterion	Strategy
A	Largest detrimental	$\hat{\beta} \geq \log(0.90)$	Harm SG; complement = benefit
B	Smallest harmful	$\hat{\beta} \geq \log(0.90)$	Min $\hat{\mathcal{H}}$; $\hat{\mathcal{H}}^c = \hat{\mathcal{G}}$
C	Largest beneficial	$\hat{\beta} \leq \log(0.60)$	Directly target strong benefit

- In **(A)** and **(B)**: identify harm $\hat{\mathcal{H}}$, then the complement $\hat{\mathcal{G}} = \hat{\mathcal{H}}^c$ is generally beneficial
- In **(C)**: directly target a clinically compelling beneficial subgroup effect

Part 1: DGM Specification & Simulation Engine



Part 2: Per-Replicate Analysis



Super population

Super-population Characteristics by Treatment Arm

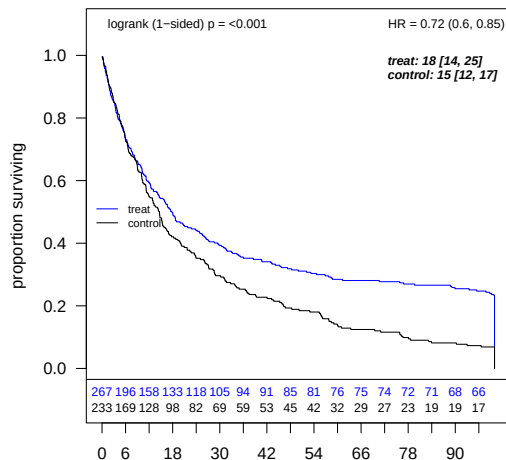
Characteristic		Control (n=2355)	Treatment (n=2645)	P-value ¹	SMD ²
z_age	Mean (SD)	60.5 (13.1)	61.0 (11.7)	0.182	0.04
z_bm	Mean (SD)	7.0 (6.4)	8.7 (7.3)	<0.001	0.25
entrytime	Mean (SD)	11.4 (4.9)	11.7 (4.7)	0.022	0.07
lin_pred_1	Mean (SD)	-0.1 (1.3)	0.2 (1.5)	<0.001	0.23
lin_pred_0	Mean (SD)	-0.2 (0.7)	-0.2 (0.7)	0.131	0.04
z_male		1674 (71.1%)	2003 (75.7%)	<0.001	0.11
z_histology		640 (27.2%)	899 (34.0%)	<0.001	0.15
z_prior_treat		1690 (71.8%)	1904 (72.0%)	0.886	0.00
z_AP		575 (24.4%)	683 (25.8%)	0.267	0.03
ecog		1257 (53.4%)	1375 (52.0%)	0.340	0.03
strat		874 (37.1%)	961 (36.3%)	0.588	0.02

¹P-values: t-test for continuous, chi-square/Fisher's exact for categorical/binary variables

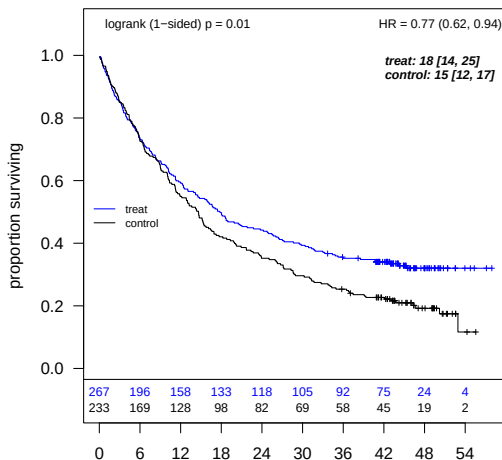
²SMD = Standardized mean difference (Cohen's d for continuous, Cramer's V for categorical)

Kaplan-Meier: ITT Ideal (No Censoring) and Censoring

ITT no censoring (through 100 months)

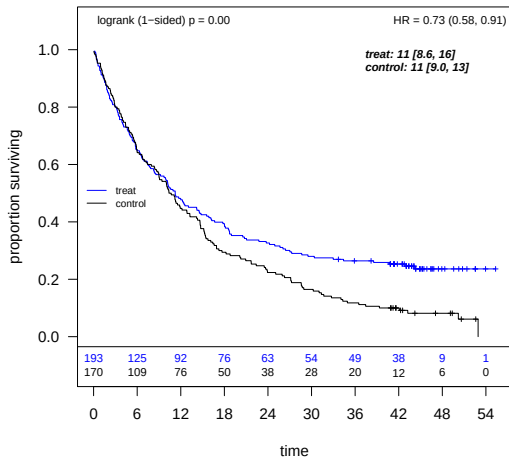


ITT censored

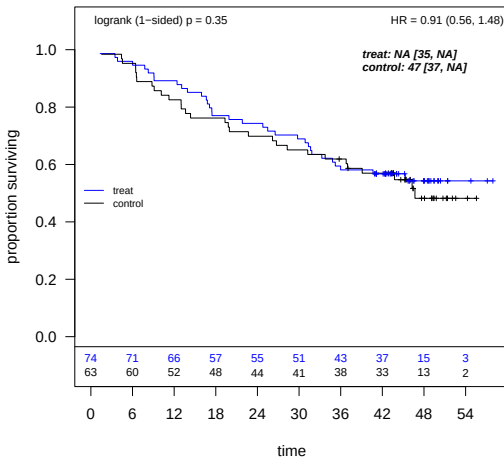


Kaplan-Meier: Non-AP and AP Populations

Non-AP (Training)

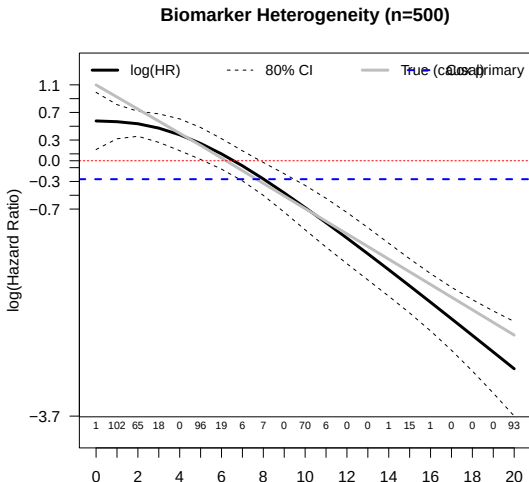


AP (Testing)



Spline Fit to Observed Data

```
cox_cs_fit(  
df = df_example,  
tte_name = "y_sim", event_name = "event_sim",  
treat_name = "treat_sim",  
truebeta_name = "loghr_po",  
z_name = "z_bm", z_by = 1, z_window = 0.5,  
alpha = 0.20,  
verbose = FALSE,  
plot_params = list(  
xlab = "Biomarker Level",  
main_title = "Biomarker Heterogeneity (n=500)"  
)
```



ForestSearch on Non-AP Data

```
Forced confounders: z_age <= 65, z_bm < 1,
  z_bm < 2, z_bm < 3, z_bm < 4, z_bm < 5,
  z_bm < 6, z_bm < 7, z_bm < 8, z_bm < 9,
  z_bm < 10
GRF subgroup: z_bm <= 3
GRF cuts identified: 3
  Cuts: strat <= 0, z_bm <= 8, z_bm <= 3
Cox-LASSO selected: 2 of 7 candidate factors
  Omitted: z_age, z_male, ecog, z_prior_treat,
    strat
Candidate factors: 19
[1] "z_bm <= 8"    "z_bm <= 3"    "z_age <= 65"  "z_bm < 1"
[5] "z_bm < 2"    "z_bm < 3"    "z_bm < 4"    "z_bm < 5"
[9] "z_bm < 6"    "z_bm < 7"    "z_bm < 8"    "z_bm < 9"
[13] "z_bm < 10"   "z_bm <= 7.7" "z_bm <= 5"   "z_bm <= 2"
[17] "z_bm <= 10"  "z_histology" "strat <= 0"
Number of possible configurations (<= maxk):
  maxk = 1 , # combinations = 38
Events criteria: control >= 10 , treatment >= 10
Sample size criteria: n >= 60
Subgroup search completed in 0 minutes

-- Filtering Summary --
Combinations evaluated: 38
Passed variance check: 38
Passed prevalence (>= 0.025 ): 37
Passed redundancy check: 36
Passed event counts (d0>= 10 , d1>= 10 ): 35
Passed sample size (n>= 60 ): 35
Cox model fit successfully: 35
Passed HR threshold (>= 0.9 ): 15
-----
Found 15 subgroup candidate(s)
```

```
# of candidate subgroups (meeting all criteria)
  = 15
Removed 7 near-duplicate subgroups
# of unique initial candidates: 8
# Restricting to top stop_Kgroups = 30
# of candidates to evaluate: 8
```

```
=====
TOP 8 CANDIDATE SUBGROUPS FOR CONSISTENCY EVALUATION
Sorted by: minSG
=====
```

Rank	HR	N	Events	K	Subgroup Definition
1	4.016	77	70	1	{z_bm < 2}
2	2.765	122	112	1	{z_bm <= 2}
3	2.682	138	128	1	{z_bm <= 3}
4	2.149	210	198	1	{z_bm < 6}
5	2.095	226	214	1	{z_bm < 7}
6	2.067	229	217	1	{z_bm < 8}
7	2.040	233	221	1	{z_bm < 10}
8	1.540	277	260	1	{z_bm <= 10}

```
-----
Parallel config: workers = 100 , batch_size = 8
Batch 1 / 1 : candidates 1 - 8
Evaluated 8 of 8 candidates (complete)
8 subgroups passed consistency threshold
SG focus = minSG
Seconds and minutes forestsearch overall =
  6.519 0.1086
Consistency algorithm used: twostage
Subgroup identified: {z_bm < 2}
```

Subgroups Identified: Non-AP Training

Identified Subgroups Single-factor subgroups (maxk=1)

Factor	N	Events	E1	HR	Pcons
{z_bm < 2}	77	70	31	4.02	1.00
{z_bm <= 2}	122	112	58	2.77	1.00
{z_bm <= 3}	138	128	63	2.68	1.00
{z_bm < 6}	210	198	104	2.15	1.00
{z_bm < 7}	226	214	116	2.10	1.00
{z_bm < 8}	229	217	119	2.07	1.00
{z_bm < 10}	233	221	123	2.04	1.00
{z_bm <= 10}	277	260	139	1.54	1.00

Search Configuration: Single-factor candidates (L) = 38; Maximum combinations evaluated = 38; Search depth (maxk) = 1

Search Results: Candidate subgroups found = 15; Maximum HR estimate = 4.02

Note: E1 = events in treatment arm; Pcons = consistency proportion

Training and Testing Estimates

Treatment Effect Estimates

Non-AP (training)

Subgroup	n	n1	events	m1	m0	RMST	HR (95% CI)	AHR(po)
ITT	363 (100.0%)	193 (53.2%)	303 (83.5%)	11.2	10.5	4.9	0.73 (0.58, 0.91)	0.76
Questionable ¹	77 (21.2%)	31 (40.3%)	70 (90.9%)	3.6	16.5	-10	4.02 (2.43, 6.64)	2.52
Recommend	286 (78.8%)	162 (56.6%)	233 (81.5%)	14.7	9.2	10	0.53 (0.41, 0.68)	0.55

¹Identified subgroup : {z_bm < 2}

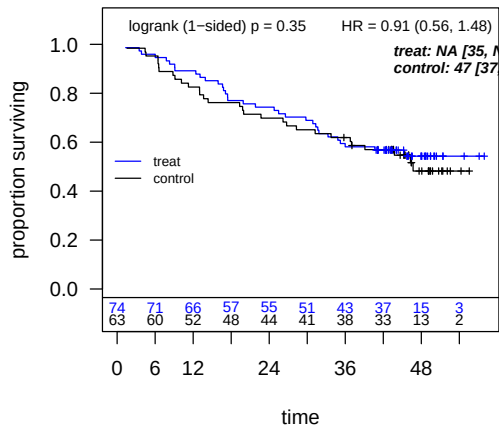
AP (testing)

Subgroup	n	n1	events	m1	m0	RMST	HR (95% CI)	AHR(po)
ITT	137 (100.0%)	74 (54.0%)	63 (46.0%)	NA	46.7	1.6	0.91 (0.56, 1.48)	0.74
Questionable ¹	26 (19.0%)	21 (80.8%)	17 (65.4%)	24.8	NA	-16	3.11 (0.89, 10.90)	2.52
Recommend	111 (81.0%)	53 (47.7%)	46 (41.4%)	NA	46.7	6.6	0.62 (0.34, 1.11)	0.55

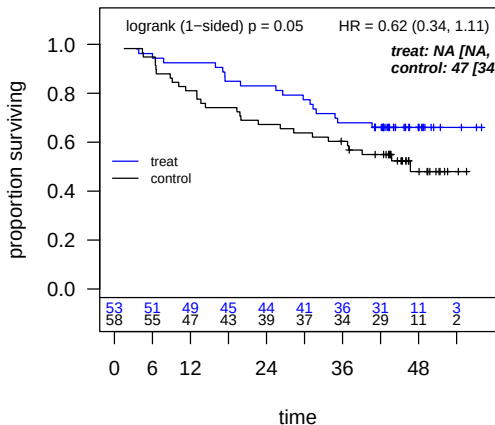
¹Identified subgroup : {z_bm < 2}

AP (ITT) vs Biomarker High (per identified ≥ 2)

AP Population

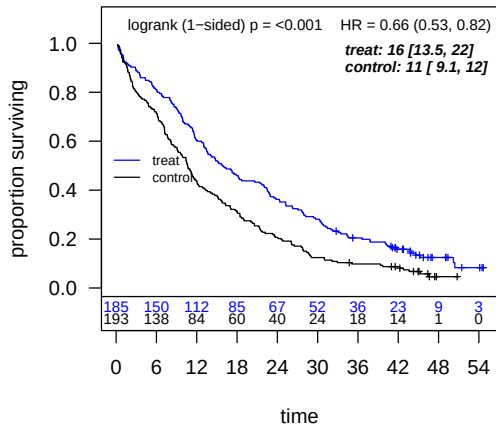


AP Population: BM ≥ 2

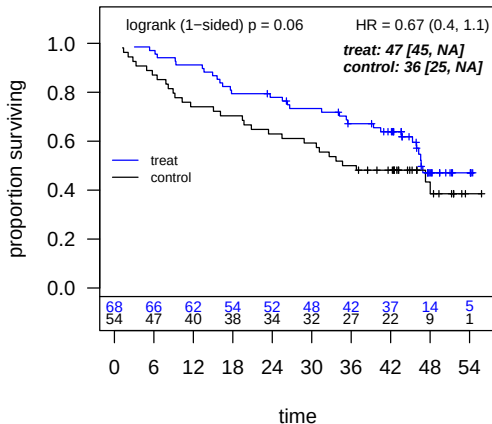


Null: Regional Kaplan-Meier Plots

Null: non-AP (Training)



Null: AP (Testing)



ForestSearch Under Null

```
Forced confounders: z_age <= 65, z_bm < 1,
  z_bm < 2, z_bm < 3, z_bm < 4, z_bm < 5,
  z_bm < 6, z_bm < 7, z_bm < 8, z_bm < 9,
  z_bm < 10
GRF: no subgroup identified
GRF cuts identified: 0
Cox-LASSO selected: 1 of 7 candidate factors
  Omitted: z_age, z_bm, z_male, ecog,
           z_histology, strat
Candidate factors: 12
[1] "z_age <= 65"      "z_bm < 1"         "z_bm < 2"
[5] "z_bm < 4"         "z_bm < 5"         "z_bm < 6"
[9] "z_bm < 8"         "z_bm < 9"         "z_bm < 10"
Number of possible configurations (<= maxk):
  maxk = 1 , # combinations = 24
Events criteria: control >= 10 , treatment >= 10
Sample size criteria: n >= 60
Subgroup search completed in 0 minutes
```

```
-- Filtering Summary --
Combinations evaluated: 24
Passed variance check: 24
Passed prevalence (>= 0.025 ): 23
Passed redundancy check: 22
Passed event counts (d0>= 10 , d1>= 10 ): 22
Passed sample size (n>= 60 ): 22
Cox model fit successfully: 22
Passed HR threshold (>= 0.9 ): 0
-----
```

NO subgroup candidates found

HTE Scenario: Simulation Setup

```
options(future.globals.onReference = "ignore")
results_alt <- mrct_region_sims(
  dgm      = dgm_spline,
  n_sims   = 5000,
  verbose_n_sims = 1L,
  n_sample = 500,
  region_var = "z_AP",

  sg_focus = "minSG", maxk = 1,
  hr.threshold = 0.90,
  hr.consistency = 0.80,
  pconsistency.threshold = 0.80,

  confounders.name = c("z_age", "z_bm",
    "z_male", "ecog", "z_histology",
    "z_prior_treat", "strat"),
  conf_force = c("z_age <= 65",
    "z_bm <= 0", "z_bm <= 1",
    "z_bm <= 2", "z_bm <= 5"),
```

```
fs_args = list(
  use_grf = TRUE, use_lasso = TRUE,
  cut_type = "default", n.min = 60,
  stop_threshold = 0.95,
  d0.min = 12, d1.min = 12,
  max_subgroups_search = 30,
  use_twostage = TRUE,
  twostage_args = list(
    n.splits.screen = 50,
    batch.size = 25,
    conf.level = 0.99)),

sim_args = list(rand_ratio = 1,
  draw_treatment = FALSE),
analysis_time = 60, cens_adjust = 0,
parallel_args = list(
  plan = "multisession",
  workers = 10, show_message = TRUE),
details = TRUE, seed = 20240101
```

)

HTE Scenario: Summary Results

Identified subgroups and estimation summaries

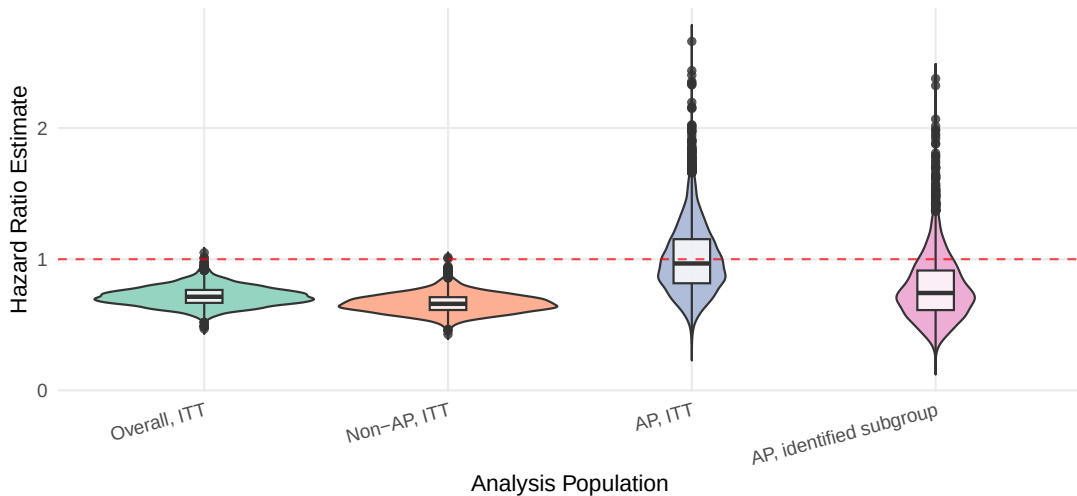
HTE: 5000 sims | Uniform: 10000 sims

Category	Metric	Scenario	
		HTE	Uniform
HR Estimates	HR ITT	0.718 (0.074)	0.735 (0.074)
	HR ITT (stratified by region)	0.707 (0.072)	0.713 (0.072)
	HR Training (non-Region A)	0.663 (0.073)	0.719 (0.079)
	HR Testing (Region A)	1.002 (0.267)	0.699 (0.198)
	HR Subgroup	0.778 (0.240)	0.698 (0.187)*
	PO HR Subgroup	0.533 (0.066)	0.700 (0.000)
Subgroup Identification	Subgroup Found	100.0%	10.7%
	Prevalence (when found)	0.816 (0.039)	0.701 (0.158)
Region A Classification	RegA HR > 0.9	3032 (60.6%)	1444 (14.4%)
	RegA(sg) HR <= 0.85	3353 (67.1%)	7956 (79.6%)
	RegA > 0.9 & RegA(sg) <= 0.85	1416 (28.3%)	25 (0.2%)
	RegA > 0.9 & RegA(sg) <= 0.80	1116 (22.3%)	16 (0.2%)
Subgroup Classification	Biomarker	4813 (96.3%)	452 (4.5%)
	Age	181 (3.6%)	274 (2.7%)

* Trimmed mean/SD: lower and upper 1% excluded (raw mean exceeded 1,000)

HTE Scenario: HR Distributions

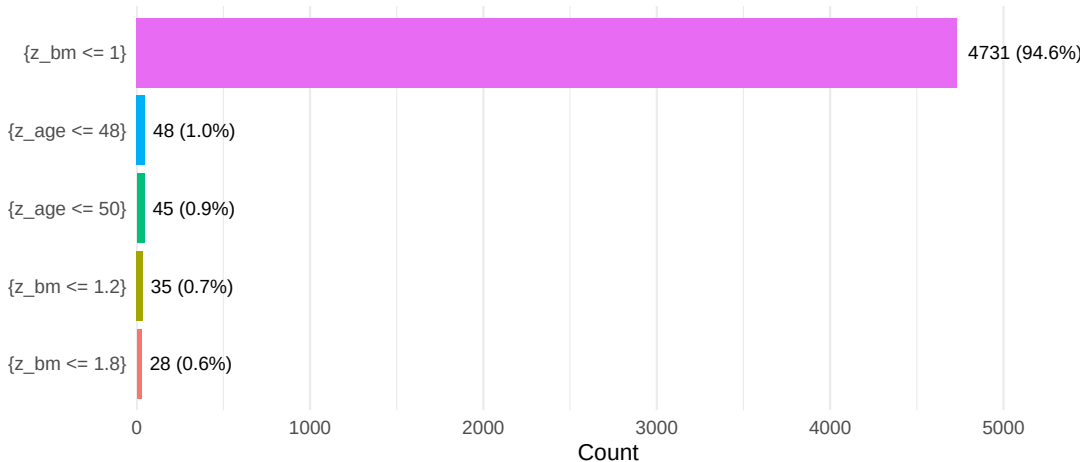
Distribution of HR Estimates Across Simulations



HTE Scenario: Subgroup Distribution

HTE Scenario: Subgroups Found

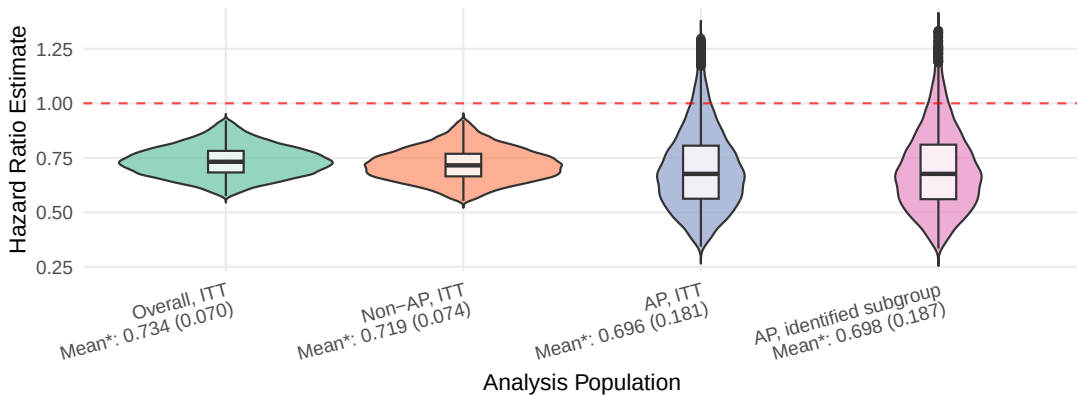
Found: 5000 / 5000 sims | Showing subgroups $\geq 0\%$ (5 of 30 distinct) | Other: 113 (2.3%)



Null Scenario: HR Distributions

Distribution of HR Estimates Across Simulations

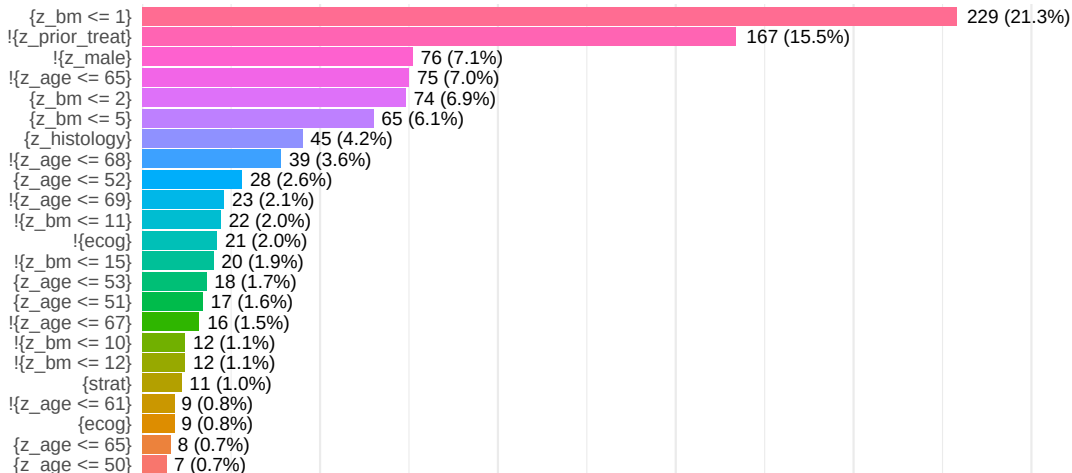
Trimmed mean/SD shown (1% symmetric trim; 800 observations flagged)



Null Scenario: Subgroup Distribution

Uniform Benefit Scenario: Subgroups Found

Found: 1074 / 10000 sims | Showing subgroups $\geq 0\%$ (23 of 62 distinct) | Other: 71 (6.6%)



Summary

- ➊ **ForestSearch** provides a principled approach to subgroup identification that directly addresses MRCT regional consistency challenges
- ➋ The **Weibull AFT/Cox dual framework** enables causal interpretation — simulated on AFT scale, interpreted via hazard ratios
- ➌ Three complementary estimands (**AHR, CDE, Marginal HR**) robustly characterize treatment effect heterogeneity
- ➍ The **train/test paradigm** (non-AP $\rightarrow$ AP) avoids overfitting
- ➎ **Data-driven simulation** evaluates full operating characteristics — false discovery, estimation bias, CI coverage — grounded in real trial data
- ➏ **Simulation results** demonstrate the approach can recover AP consistency under heterogeneous effects while controlling false-discovery under the null

Thank You

ForestSearch R Package

Leon, L.F., Jemielita, T., Guo, Z., Marceau West, R. & Anderson, K.M. (2024)

Exploratory subgroup identification in the heterogeneous Cox model.

Statistics in Medicine. doi:10.1002/sim.10163

```
remotes::install_github("larry-leon/forestsearch")
```

<https://larry-leon.github.io/forestsearch/>

R code throughout this deck is directly executable with the ForestSearch package and the synthetic case-study dataset.

Some More Details

Dual Causal Status

The Weibull AFT scale-change parameter $\exp(-\beta)$ equals the **geometric mean ratio** of potential survival times:

$$\exp(E[\log T_i(1) - \log T_i(0)])$$

Simultaneously **individual-level** and **population-level** causal.

AHR Inherits This Status:

$$\text{AHR}(\mathcal{S}) = \exp\left(\frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} [\theta_i(1) - \theta_i(0)]\right)$$

Cox HR as Operational Tool

- *True* effects: defined through $\psi^0(z)$
- Subgroup *selection*: Cox HR as efficient screening
- Subgroup *evaluation*: AHR and CDE
- *Correction*: bootstrap bias correction (hr.H.bc)

De-Biased Estimation:

$$\hat{\beta}^*(\hat{\mathcal{H}}) = \hat{\beta}(\hat{\mathcal{H}}) - \frac{1}{B} \sum_{b=1}^B [\eta_b^*(\hat{\mathcal{H}}_b^*) + \eta_b^*(\hat{\mathcal{H}})]$$

Infinitesimal jackknife for valid CIs.

ForestSearch Implementation Details

Feature Discovery Pipeline

- ① **GRF** (causal survival forests): treatment effect heterogeneity via RMST
- ② **LASSO** regularization: variable screening via Cox penalized regression
- ③ **Exhaustive combinatorial search**: across factors and cutpoints
- ④ **Forced cutpoints**: `conf_force` for domain-guided biomarker cuts

Selection Criteria

- HR threshold: $\hat{\beta} \geq \log(0.90)$
- Consistency rate $\geq 80\text{--}90\%$
- Minimum sizes: $n \geq 60$, $d \geq 12$
- `sg_focus`: "minSG" | "hr" | "maxSG"

Cross-Validation

- leave-one-out
- 10-fold
- agreement metrics: cv-testing vs full sens, ppv, subgroup match (cv-training vs full) metrics

Bootstrap Inference

- parallelised (dofuture) bootstrap
- Full algorithm mimicked in each replicate ($B = 300\text{--}2000$)
- IJ variance estimation for valid confidence intervals